MILITARY SPECIFICATION

SUPPORTS, MOLDED, PLASTIC (FOR USE IN AMMUNITION PACKAGING)

1. SCOPE

1.1 Scope. This specification covers colored parts, molded from high density polyethylene, and used as supports for standard contour and various types of ammunition fuzes within shipping containers.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons; General Specification for

STANDARDS

FEDERAL

FED-STD-595 - Colors

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105).
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1168 - Lot Numbering of Ammunition

FSC: 8140

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MIL-STD-1169 - Packaging, Packing and Marking for Shipping of Inert Ammunition Components; General Specification for
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U. S. ARMY MUNITIONS COMMAND

9232148 - Support, Bottom
9232149 - Support, Top

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS PUBLICATIONS

D1238-70 - Measuring Flow Rate of Thermoplastics by Extrusion Plastometer
D1505-68 - Density of Plastics by the Density Gradient Technique

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadlephia, Pennsylvania, 19103).

3. REQUIREMENTS

3.1 Material. The resin used in the manufacture of the supports shall be high density polyethylene or copolymer thereof, which contains 0.3 percent, plus or minus 0.05 percent carbon black, an anti-oxidant uniformly dispersed. In addition, the supports shall be in accordance with the applicable drawings and specifications.

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3.2 Physical properties. The high density polyethylene supports shall conform to the physical properties listed in Table I when tested as specified in 4.4.

TABLE I
PHYSICAL PROPERTIES

Test	Requirements
Melt Index, grams/10 minutes, with 2160g load (Condition E)	3.0 max.
Density, gms/cc	0.955 min.
Color Number (FED-STD-595)	All sections of the supports shall closely approximate composition color numbers 14036, 14052, 14056, 14062, 14064 or 14077. Although specified colors are listed as gloss, the surface finish on the supports will be as produced from good molding practices.

3.3 Workmanship. All parts shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of amides, water, dirt, twists, and scrap. The cleaning method shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents.

3.4 First article inspection. This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where

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such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Submission of product. At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.

b. Specification number and date, together with an identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, wher such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Inspection (see 6.1)

4.2.1 Submission. Prior to initiation of regular production the contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2. The first article sample shall consist of twenty-five complete sets of parts from each die or mold which have been produced by the contractor or furnished by a supplier which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of

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this inspection with the sample, including statements of findings for materials, processes and tests. A sample containing known defects will not be submitted unless specifically authorized by the contracting officer. A first article sample, or portion thereof, as directed by the contracting officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specification such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be performed. Supports shall be subjected by the Government to any or all of the inspections and tests specified in this specification. In addition, one support from each cavity of each die or mold, as applicable, shall be inspected for compliance with any or all of the requirements specified or the applicable drawing.

4.2.3 Rejection. If any support fails to comply with any or all of the applicable requirements, the first article sample shall be rejected. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and subrit new or supplemental quantities until such time as an acceptable sample is submitted, or until otherwise directed by the contracting officer.

4.3 Inspection provisions

4.3.1 Lot formation. The term "inspection lot", as used in this specification, is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process, in accordance with the same drawing and same drawing revision, same specification and same specification revision, and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process or specification, not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein,

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regardless of the type of sampling procedure which is being applied to determined conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each lot shall contain:

a. Resin from one lot from the same manufacturer.
b. Resin of one color designation.

4.3.2 Examination. The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.2.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspection listed shall be in accordance with 4.3.4.

4.3.2.1 Support, Top (see dwg. 9232149)

Categories	Defects	Method of Inspection (CD)	Code No. (see 6.2)
Critical: None defined.			
Major:	AQL 0.65 percent		
101.	Wall thickness	Gage	01001
102.	Cavity mislocated	Gage	01016
103.	Crack	Visual	01002
Minor:	AQL 4.0 percent		
201.	Large inside diameter (between ribs) of support	Gage	01003
202.	Small inside diameter (between ribs) of support	Gage	01004
203.	Total length	Gage	01005
204.	Total width	Gage	01006

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205.	Total height	Gage	01007
206.	Width of support rib	Gage	01008
207.	Thickness of support rib	Gage	01009
208.	Width of web	Gage	01010
209.	Height of web	Gage	01011
210.	Location of inner shoulder from rearward face	Gage	01012
211.	Hole missing	Visual	01013
212.	Evidence of poor workmanship (see 3.3)	Visual	01014
213.	Marking misleading, missing or unidentifiable	Visual	01015

4.3.2.2 Support, Bottom (see dwg. 9232148)

Categories	Defects	Method Inspection (CD)	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Wall thickness	Gage	02001
102.	Cavity mislocated	Gage	02014
103.	Crack	Visual	02002
Minor:	AQL 4.0 percent		
201.	Total length	Gage	02003
202.	Total width	Gage	02004
203.	Large internal diameter of fuze cavity	Gage	02005
204.	Small internal diameter of fuze cavity	Gage	02006
205.	Location of shoulder (within fuze cavity) from forward face	Gage	02007
206.	Total depth of fuze cavity	Gage	02008
207.	Cavity cut-out mislocated	Gage	02009
208.	Total height	Gage	02010
209.	Hole missing	Visual	02011
210.	Evidence of poor workmanship (see 3.3)	Visual	02012
211.	Marking misleading, missing or unidentifiable	Visual	02013

4.3.3 Tests

4.3.3.1 Classification of tests. The examination and testing of the supports shall be classified as follows: (see 6.5)

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a. First Article Testing
b. Regular Production Tests

4.3.3.1.1 First article tests.The first article tests shall consist of all inspections, tests, and quantities as specified in 4.2 and 4.3, using test equipment specified in 4.3.4.

4.3.3.1.2 Regular production tests.The regular production test for acceptance of individual lots shall consist of inspections and tests as specified in 4.3.4.

4.3.3.2 Delay time in dimensioning examinations. No attempt shall be made to determine compliance with dimension and tolerance requirements of the sections for a minimum period of 48 hours after molding.

4.3.3.3 Melt Index. (see Table I), Major defect, Code No. 03001. Three determinations shall be made and averaged from each lot for the test. Failure of average to meet the values when tested as specified in 4.4.1 shall be cause for rejection of the lot.

4.3.3.4 Density (see Table I), Major defect, Code No. 04001. Three determinations shall be made and averaged from each lot for this test. Failure of average to meet the values when tested as specified in 4.4.2 shall be cause for rejection of the lot.

4.3.4 Inspection equipment. Inspection equipment required to perform the inspection prescribed in this specification is identified in the Examination (4.3.2) and Test Methods (4.4) paragraphs herein. See 6.3 for details concerning responsibilities for inspection equipment design and approval.

4.4 Test methods and procedures

4.4.1 Melt index. Melt index shall be determined in accordance with ASTM D-1238-70. The test specimen shall be cut from three supports.

4.4.2 Density. The density of the specimens shall be determined in accordance with ASTM D-1505-68. The test specimen shall be cut from three supports. Specimens shall not be annealed or otherwise treated so as to change the cyrstallinity (see 6.6).

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5. PREPARATION FOR DELIVERY

5.1 Packing

5.1.1 Level A and C. Supports, suitably protected from contamination, shall be packed in accordance with applicable Level of Standard MIL-STD-1169.

5.2 Marking

5.2.1 Level A and C. Shipments shall be marked in accordance with Standard MIL-STD-1169.

6. NOTES

6.1 Ordering data. Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Packing required (see 5.1)
c. Provisions for submission of First Article Sample

6.2 Inspection code numbers. The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. In addition, they should be cited as references on drawings of equipment designs submitted by the contractor to the Government for approval (see 6.3.3).

6.3 Inspection Equipment Designs. Inspection equipment designs are of two types-Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers (on the Equipment Lists referenced on the Equipment Tabulation) or (under the "Method of Inspection" headings in Section 4). Design responsibility for all other inspection equipment is assigned to the contractor. Equipment to be designed by the contractor is designated "CD". The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

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6.3.1 SIE designs. SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detail design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor designs. Contractor designs are required for all inspection equipment for which SIE design are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public.) Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commerical equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.3.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five digit defect code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design together with the component or assembly number and revision letter to which the specific design applies.

6.3.3 Submission of designs for approval. Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval to Picatinny Arsenal, SARPA-QA-A-P, Dover, New Jersey, 07801 in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final

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submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.4 Molding instructions. The particular material selected as to density and melt index should be injection molded so as to meet the requirements of this specification. It must be noted that selection of a material complying with the density and melt index requirements do not necessarily insure compliance with the performance requirements of the specification nor relieve the molder of his responsibility for molding in such a fashion that the supports will meet the requirement.

6.5 Classification of tests. The term classification of tests as used in the text of this specification are intended to be used as follows:

6.5.1 First article tests will be the examinations and tests performed prior to the start of regular production to assure the Government that the methods and processes being used for production will produce an end product which will be acceptable to the Government under this specification.

6.5.2 Regular production tests will be the examination and tests performed on a lot to lot basis during regular production to assure the Government that the methods and processes being used for production will produce an end product which will be acceptable under this specification.

6.6 Density specimen. Saw speed in excess of 500 to 600 lineal feet per minute have been found to be unsatisfactory as excess speed can essentially melt the material and greatly change the density.

6.7 Temperatures for continuous use or storage of the molded supports should not exceed 176 degrees F.

6.8 Intended use. The molded supports covered by this specification are intended for use in supporting standard contour and various types of ammunition fuzes within shipping containers.

Custodian:
ARMY-MU

Preparing Activity:
ARMY-MU

Project Number: 8140-A115

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No 22-R255

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CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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